

Form Number : _____

Paper Code : 3002CM370218010

English



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2019 – 2020)

NEET : PRACTICE PAPER

TARGET : PRE-MEDICAL - 2020

Test Type : PT # 04

Test Pattern : NEET

TEST DATE : 06-04-2020

Important Instructions

Do not open this Test Booklet until you are asked to do so

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is **3 Hours** and Questions Paper Contains **180** Questions. The Max. Marks are **720**.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while **1 mark will be deducted for every wrong answer**. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of **Blue / Black Ball Point Pen only** as the correct answer(s) of the question attempted.
8. **Use of Pencil is strictly prohibited.**

Note : In case of any Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with **Paper code** and Your **Form No.**

Your Target is to secure Good Rank in Pre-Medical

Corporate Office : **ALLEN** CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-5156100 ✉ info@allen.ac.in 🌐 www.allen.ac.in

TOPIC COVER

PHYSICS : Electrostatics + Capacitance

CHEMISTRY : Phenol, Ether + Aldehyde, Ketones + Carboxylic acid & derivatives + Amines & Anilene

BIOLOGY : Plant Physiology (Complete) + Animal Tissue + Digestive system + Respiratory system + Excretory system + Circulatory system + Taxonomy , Biological Classification

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

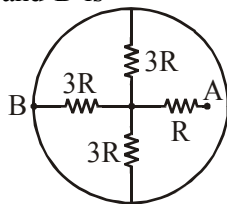
BEWARE OF NEGATIVE MARKING

1. A nonconducting ring of radius R has charge Q distributed unevenly over it. If it rotates with an angular velocity ω about an axis passing through center and perpendicular to the plane of ring, the equivalent current will be

(1) Zero (2) $Q\omega$ (3) $Q\frac{\omega}{2\pi}$ (4) $Q\frac{\omega}{2\pi R}$

2. In the network shown below, the ring has zero resistance. The equivalent resistance between the point A and B is

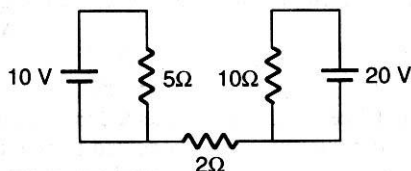
(1) $2R$
(2) $4R$
(3) $7R$
(4) $10R$



3. A wire of resistance R is bent in the form of circle. The resistance between diametrically opposite point is

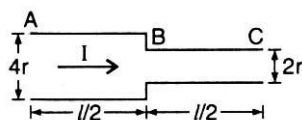
(1) $R/2$ (2) $2R$
(3) $R/4$ (4) None of these

4. Find the current through 2Ω



(1) 0 (2) 1 amp (3) 2 amp (4) 4 amp

5. Consider a cylindrical element as shown in the adjoining figure. Current flowing through the element is I and resistivity of the material of the cylinder is ρ . Choose the correct option out of the following.



- (1) Resistance of second half is four times that of first half
(2) Voltage drop in first half is twice of voltage drop in second half
(3) Current density in both halves are equal
(4) Electric field in both halves is equal

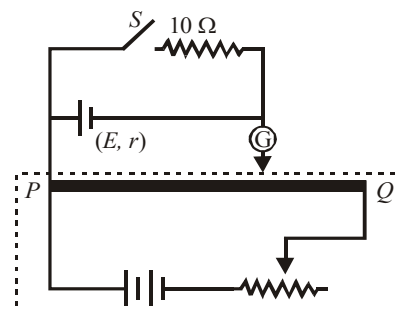
6. Current starts flowing through a conductor at $t = 0$ is given by $I = 8 - 2t^2$ (where ' t ' is time). Find the amount of charge flow inside the conductor before the current started reversing its direction.

(1) $\frac{8}{3}$ unit (2) $\frac{16}{3}$ unit
(3) $\frac{32}{3}$ unit (4) 16 unit

7. A wire of length L is drawn such that its diameter is reduced to half of its original diameter. If the initial resistance of the wire were 10Ω , its new resistance would be :

(1) 40Ω (2) 80Ω (3) 120Ω (4) 160Ω

8. The dotted box is a potentiometer circuit. In the driving circuit, if switch is open balancing point is at 60 cm from P and when switch is closed, the balancing point is 50 cm from P . Then internal resistance of cell is

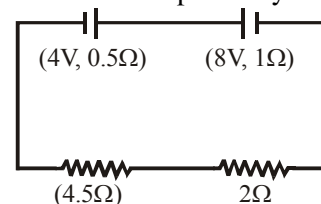


(1) 0.5Ω (2) 2Ω (3) 2.5Ω (4) 4Ω

9. The total power dissipated by a cell (E, r) , when it dissipates maximum power across an external resistance is

(1) $\frac{2E^2}{r}$ (2) $\frac{E^2}{r}$ (3) $\frac{E^2}{2r}$ (4) $\frac{E^2}{4r}$

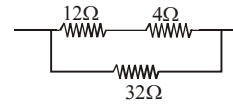
10. In the circuit, potential difference across 4V and 8V cells are respectively



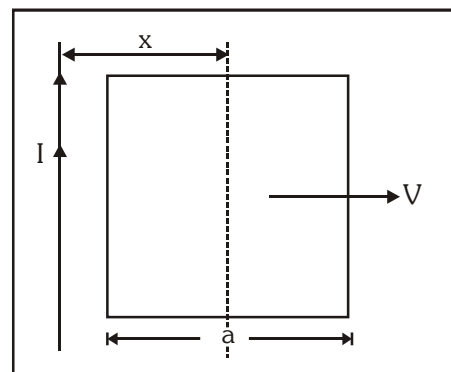
(1) 3.75 V, 7.5 V (2) 4.25 V, 7.5 V
(3) 4.25 V, 8.5 V (4) 3.75 V, 8.5 V

11. If E is the emf of a cell and V is the terminal voltage across it, then match the following
- Discharging circuit (i) $V = E$
 - Charging circuit (ii) $V > E$
 - Open circuit (iii) $V < E$
 - Short circuit (iv) $V = 0$
- a(iii), b(ii), c(iv), d(i)
 - a(ii), b(iv), c(i), d(iii)
 - a(iii), b(ii) c(i), d(iv)
 - a(iii), b(iv), c(i), d(ii)
12. In a mixed grouping circuit, 640 number of identical cells each of (20 V, 1.5Ω) are connected such that x number of cells are in series in a row and there are y such rows in parallel to each other across an external resistance of 15Ω , to get maximum current in external resistor
- $x = 40, y = 16$
 - $x = 80, y = 8$
 - $x = 8, y = 80$
 - $x = 16, y = 40$
13. Two similar heater coils separately take 10 min to boil a certain amount of water when connected across same source. If both coils are connected in series across same source, time taken to boil the same amount of water will be
- 15 min
 - 20 min
 - 7.5 min
 - 25 min
14. An electric bulb is rated 220 volt -100 watt. The power consumed by it when operated on 110 volt will be
- 50 watt
 - 75 watt
 - 40 watt
 - 25 watt

15. If power dissipated in the 32Ω resistor is 10 cal/sec, the power dissipated in 12Ω resistor is

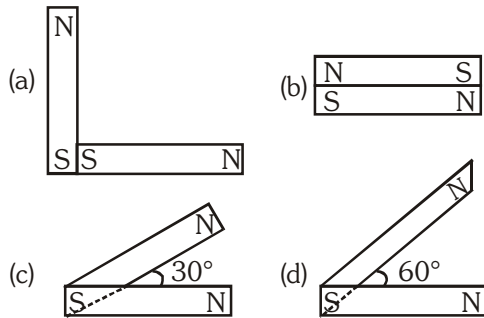


- 20 cal/sec
 - 10 cal/sec
 - 15 cal/sec
 - 5 cal/sec
16. A resistance 'R' draws power 'P' when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes 'Z', the power drawn will be:
- $P \sqrt{\frac{R}{Z}}$
 - $P \left(\frac{R}{Z} \right)$
 - P
 - $P \left(\frac{R}{Z} \right)^2$
17. A conducting square frame of side 'a' and a long straight wire carrying current I are located in the same plane as shown in the figure. The frame moves to the right with a constant velocity 'V'. The emf induced in the frame will be proportional to :



- $\frac{1}{(2x - a)^2}$
- $\frac{1}{(2x + a)^2}$
- $\frac{1}{(2x - a)(2x + a)}$
- $\frac{1}{x^2}$

18. Following figures show the arrangement of bar magnets in different configurations. Each magnet has magnetic dipole moment $\vec{m}$. Which configuration has highest net magnetic dipole moment ?

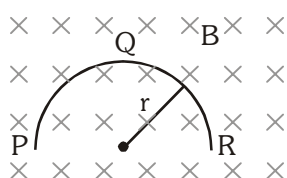


- (1) (a) (2) (b) (3) (c) (4) (d)

19. Two identical long conducting wires AOB and COD are placed at right angle to each other, with one above other such that 'O' is their common point for the two. The wires carry I_1 and I_2 currents respectively. Point 'P' is lying at distance 'd' from 'O' along a direction perpendicular to the plane containing the wires. The magnetic field at the point 'P' will be :-

- (1) $\frac{\mu_0}{2\pi d} \left(\frac{I_1}{I_2} \right)$ (2) $\frac{\mu_0}{2\pi d} (I_1 + I_2)$
(3) $\frac{\mu_0}{2\pi d} (I_1^2 - I_2^2)$ (4) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)^{1/2}$

20. A thin semicircular conducting ring (PQR) of radius 'r' is falling with its plane vertical in a horizontal magnetic field B, as shown in figure. The potential difference developed across the ring when its speed is v, is :-

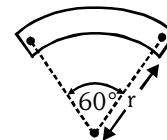


- (1) Zero
(2) $Bv\pi r^2 / 2$ and P is at higher potential
(3) $\pi r B v$ and R is at higher potential
(4) $2r B v$ and R is at higher potential

21. A current loop in a magnetic field :-

- (1) Can be in equilibrium in two orientations, one stable while the other is unstable.
(2) Experiences a torque whether the field is uniform or non uniform in all orientations
(3) Can be in equilibrium in one orientation
(4) Can be in equilibrium in two orientations, both the equilibrium states are unstable

22. A bar magnet of length ' ℓ ' and magnetic dipole moment 'M' is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be



- (1) $\frac{M}{2}$ (2) M (3) $\frac{3}{\pi} M$ (4) $\frac{2}{\pi} M$

23. A coil of self-inductance L is connected in series with a bulb B and an AC source. Brightness of the bulb decreases when :

- (1) an iron rod is inserted in the coil.
(2) frequency of the AC source is decreased.
(3) number of turns in the coil is reduced.
(4) A capacitance of reactance $X_C = X_L$ is included in the same circuit.

24. When a proton is released from rest in a room, it starts with an initial acceleration a_0 towards west. When it is projected towards north with a speed v_0 it moves with an initial acceleration $3a_0$ towards west. The electric and magnetic fields in the room are:

- (1) $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ down
(2) $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ up
(3) $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ down
(4) $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ up

25. A wire loop is rotated in magnetic field. The frequency of change of direction of the induced e.m.f. is :

- (1) Six times per revolution
- (2) Once per revolution
- (3) twice per revolution
- (4) four times per revolution

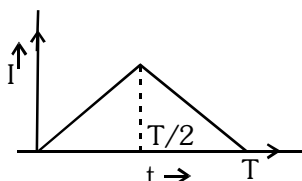
26. A coil of resistance 400Ω is placed in a magnetic field. If the magnetic flux ϕ (Wb) linked with the coil varies with time t (sec) as

$$\phi = 50t^2 + 4.$$

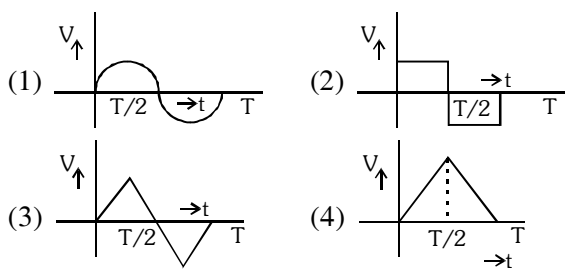
The current in the coil at $t = 2$ sec is :

- (1) 2A (2) 1A (3) 0.5A (4) 0.1A

27. The current (I) in the inductance is varying with time according to the plot shown in figure.



Which one of the following is the correct variation of voltage with time in the coil ?



28. In an electrical circuit R , L , C and an a.c. voltage source are all connected in series. When L is removed from the circuit, the phase difference between the voltage and the current in the circuit is $\pi/3$. If instead, C is removed from the circuit the phase difference is again $\pi/3$. The power factor of the circuit is :

- (1) 1 (2) $\sqrt{3}/2$ (3) $\frac{1}{2}$ (4) $\frac{1}{\sqrt{2}}$

29. A compass needle which is allowed to move in a horizontal plane is taken to a geomagnetic pole. It:

- (1) will stay in north-south direction only
- (2) will stay in east-west direction only
- (3) will become rigid showing no movement
- (4) will stay in any position

30. Two similar coils of radius R are lying concentrically with their planes at right angles to each other. The currents flowing in them are I and $2I$ respectively. The resultant magnetic field induction at the centre will be :

- (1) $\frac{\mu_0 I}{2R}$ (2) $\frac{\mu_0 I}{R}$
- (3) $\frac{\sqrt{5}\mu_0 I}{2R}$ (4) $\frac{3\mu_0 I}{2R}$

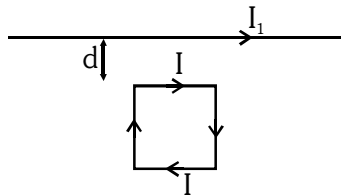
31. An alternating electric field, of frequency ν , is applied across the dees (radius = R) of a cyclotron that is being used to accelerate protons (mass = m). The operating magnetic field (B) used in the cyclotron and the kinetic energy (K) of the proton beam, produced by it, are given by :

- (1) $B = \frac{2\pi m\nu}{e}$ and $K = 2m\pi^2\nu^2 R^2$
- (2) $B = \frac{m\nu}{e}$ and $K = m^2\pi\nu R^2$
- (3) $B = \frac{m\nu}{e}$ and $K = 2m\pi^2\nu^2 R^2$
- (4) $B = \frac{2\pi m\nu}{e}$ and $K = m^2\pi\nu R^2$

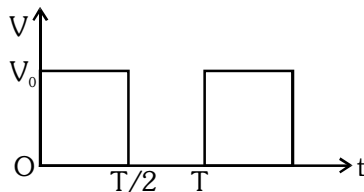
32. Charge q is uniformly spread on a thin ring of radius R . The ring rotates about its axis with a uniform frequency f Hz. The magnitude of magnetic induction of the center of the ring is

- (1) $\frac{\mu_0 q f}{2\pi R}$ (2) $\frac{\mu_0 q f}{2R}$
- (3) $\frac{\mu_0 q}{2fR}$ (4) $\frac{\mu_0 q}{2\pi fR}$

33. A square loop, carrying a steady current I , is placed in a horizontal plane near a long straight conductor carrying a steady current I_1 at a distance d from the conductor as shown in figure. The loop will experience :-



- (1) A net attractive force towards the conductor
(2) A net repulsive force away from the conductor
(3) A net torque acting upward perpendicular to the horizontal plane
(4) A net torque acting downward normal to the horizontal plane
34. The r.m.s. value of potential difference V shown in the figure is :-



- (1) $\frac{V_0}{\sqrt{3}}$ (2) V_0
(3) $\frac{V_0}{\sqrt{2}}$ (4) $\frac{V_0}{2}$
35. A coil has resistance 30 ohm and inductive reactance 20 Ohm at 50 Hz frequency. If an ac source, of 200 volt, 100 Hz, is connected across the coil, the current in the coil will be
- (1) 2.0 A (2) 4.0 A
(3) 8.0 A (4) $\frac{20}{\sqrt{13}}$ A
36. The dimensions of $(\mu_0 \epsilon_0)^{-\frac{1}{2}}$ are :-
- (1) $[L^2 T^{-2}]$ (2) $[L^{-1} T]$
(3) $[L T^{-1}]$ (4) $[L^{-\frac{1}{2}} T^{\frac{1}{2}}]$

37. An ac voltage is applied to a resistance R and an inductor L in series. If R and the inductive reactance are both equal to 3Ω , the phase difference between the applied voltage and the current in the circuit is :-

(1) $\pi/6$ (2) $\pi/4$ (3) $\pi/2$ (4) Zero

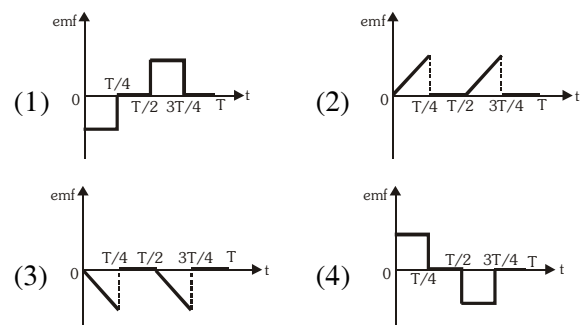
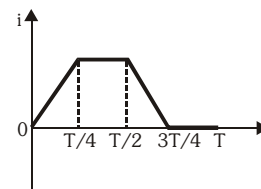
38. In an ac circuit an alternating voltage $e = 200\sqrt{2} \sin 100t$ volts is connected to a capacitor of capacity $1\mu F$. The r.m.s. value of the current in the circuit is :-

(1) 10 mA (2) 100 mA
(3) 200 mA (4) 20 mA

39. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electrons :

(1) will turn towards right of direction of motion
(2) speed will decrease
(3) speed will increase
(4) will turn towards left of direction of motion.

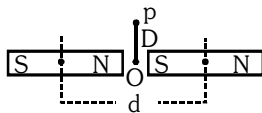
40. The current i in a coil varies with time as shown in the figure. The variation of induced emf with time would be :



41. A current loop consists of two identical semicircular parts each of radius R , one lying in the x - y plane and the other in x - z plane. If the current in the loop is i . The resultant magnetic field due to the two semicircular parts at their common centre is :-

- (1) $\frac{\mu_0 i}{2R}$ (2) $\frac{\mu_0 i}{4R}$
(3) $\frac{\mu_0 i}{\sqrt{2}R}$ (4) $\frac{\mu_0 i}{2\sqrt{2}R}$

42. Two identical bar magnets are fixed with their centres at a distance d apart. A stationary charge Q is placed at P in between the gap of the two magnets at a distance D from the centre O as shown in the figure. The force on the charge Q is :-



- (1) Directed along OP
(2) Directed along PO
(3) Directed perpendicular to the plane of paper
(4) zero

43. A closely wound solenoid of 2000 turns and area of cross-section $1.5 \times 10^{-4} \text{ m}^2$ carries a current of 2.0 A . It is suspended through its centre and perpendicular to its length, allowing it to turn in a horizontal plane in a uniform magnetic field $5 \times 10^{-2} \text{ Tesla}$ making an angle of 30° with the axis of the solenoid. The torque on the solenoid will be :-

- (1) $1.5 \times 10^{-3} \text{ N.m}$
(2) $1.5 \times 10^{-2} \text{ N.m}$
(3) $3 \times 10^{-2} \text{ N.m}$
(4) $3 \times 10^{-3} \text{ N.m}$

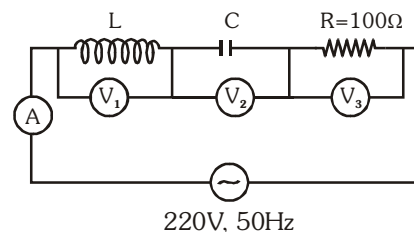
44. A particle having a mass of 10^{-2} kg carries a charge of $5 \times 10^{-8} \text{ C}$. The particle is given an initial horizontal velocity of 10^5 ms^{-1} in the presence of electric field $\vec{E}$ and magnetic field $\vec{B}$. To keep the particle moving in a horizontal direction, it is necessary that :-

- (a) $\vec{B}$ should be perpendicular to the direction of velocity and $\vec{E}$ should be along the direction of velocity
(b) Both $\vec{B}$ and $\vec{E}$ should be along the direction of velocity
(c) Both $\vec{B}$ and $\vec{E}$ are mutually perpendicular and perpendicular to the direction of velocity
(d) $\vec{B}$ should be along the direction of velocity and $\vec{E}$ should be perpendicular to the direction velocity

Which one of the following pairs of statements is possible. ?

- (1) (c) and (d) (2) (b) and (c)
(3) (b) and (d) (4) (a) and (c)

45. In the given circuit the reading of voltmeter V_1 and V_2 are 300 volts each. The reading of the voltmeter V_3 and ammeter A are respectively :



- (1) 100 V, 2.0 A
(2) 150 V, 2.2 A
(3) 220 V, 2.2 A
(4) 220 V, 2.0 A

46. E° of Mg^{2+}/Mg , Zn^{2+}/Zn and Fe^{2+}/Fe are -2.37 V, -0.76 V and -0.44 V respectively which of the following is correct:
- Mg oxidises Zn
 - Zn oxidises Fe
 - Zn reduces Mg^{2+}
 - Zn reduces Fe^{2+}
47. A compound contains element X, Y, Z with O no. +3, +5 and -2 respectively. What will be possible formula of compound containing above element.
- XYZ
 - $\text{Y}_2(\text{XZ}_3)_2$
 - $\text{X}_3(\text{YZ}_4)_3$
 - $\text{X}_3(\text{Y}_4\text{Z})_2$
48. How many mole of FeSO_4 , $\text{H}_2\text{C}_2\text{O}_4$ and FeC_2O_4 are oxidised separately by one mole of KMnO_4 in acidic medium respectively :-
- 5 ; $5/2$; $5/3$
 - $2/5$; 2 ; 5
 - $5/3$; $3/5$; $2/5$
 - $3/5$; 2 ; 5
49. Which of the following equations is a balanced one:-
- $5\text{BiO}_3^- + 22\text{H}^+ + \text{Mn}^{2+} \rightarrow 5\text{Bi}^{3+} + 7\text{H}_2\text{O} + \text{MnO}_4^-$
 - $5\text{BiO}_3^- + 14\text{H}^+ + 2\text{Mn}^{2+} \rightarrow 5\text{Bi}^{3+} + 7\text{H}_2\text{O} + 2\text{MnO}_4^-$
 - $2\text{BiO}_3^- + 4\text{H}^+ + \text{Mn}^{2+} \rightarrow 2\text{Bi}^{3+} + 2\text{H}_2\text{O} + \text{MnO}_4^-$
 - $6\text{BiO}_3^- + 12\text{H}^+ + 3\text{Mn}^{2+} \rightarrow 6\text{Bi}^{3+} + 6\text{H}_2\text{O} + \text{MnO}_4^-$
50. The molarity of a 0.2N Na_2CO_3 solution will be
- 0.05 M
 - 0.2 M
 - 0.1 M
 - 0.4 M
51. 1.8 gm of fructose is added to 2 kg of water. The freezing point of the solution is ($K_f = 1.86 \text{ K molality}^{-1}$) :-
- -1.86°C
 - 0.0093°C
 - -0.0186°C
 - -0.0093°C
52. The vapour pressure (V.P.) of a dilute solution of non-volatile solute is P and the V.P. of pure solvent is P° , the lowering of the V.P. is.
- +ve
 - ve
 - P/P°
 - P°/P
53. The Van't Hoff factor i for a compound which undergoes dissociation in one solvent and association in other solvent is respectively.
- Less than one and greater than one
 - Less than one and less than one
 - Greater than one and less than one
 - Greater than one and greater than one
54. A solute in trimerise in benzene. The minimum value of Van't Hoff factor is
- 1
 - $\frac{1}{2}$
 - $\frac{1}{3}$
 - 3
55. When equimolar aqueous solutions of glucose sodium chloride and barium nitrate are compared the vapour pressure of the solutions will be in the following order.
- Glucose > NaCl > $\text{Ba}(\text{NO}_3)_2$
 - Glucose = NaCl = $\text{Ba}(\text{NO}_3)_2$
 - $\text{Ba}(\text{NO}_3)_2$ > NaCl > Glucose
 - NaCl > $\text{Ba}(\text{NO}_3)_2$ > Glucose
56. y gm. of a non-volatile substance (molecular mass M) is dissolved in 250 gm. of benzene. If K_b is the molal elevation constant, the value of ΔT_b is given by :-
- $4M/K_b \cdot y$
 - $4 K_b \cdot y/M$
 - $K_b \cdot y/4M$
 - $K_b \cdot y/M$
57. In acidic medium, reaction $\text{MnO}_4^- \rightleftharpoons \text{Mn}^{2+}$ is an example of -
- Oxidation by three electrons
 - Reduction by three electrons
 - Oxidation by five electrons
 - Reduction by five electrons
58. The oxidation number of phosphorus in $\text{Ba}(\text{H}_2\text{PO}_2)_2$ is-
- +3
 - +2
 - +1
 - 1
59. In the following unbalanced redox reaction, $\text{Cu}_3\text{P} + \text{Cr}_2\text{O}_7^{2-} \longrightarrow \text{Cu}^{2+} + \text{H}_3\text{PO}_4 + \text{Cr}^{3+}$ Equivalent weight of H_3PO_4 is -
- $\frac{M}{3}$
 - $\frac{M}{6}$
 - $\frac{M}{7}$
 - $\frac{M}{8}$

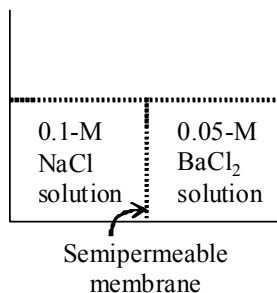
60. Oxidation number of Fe in Fe_3O_4 is fractional because-
- It is a mixed $[\text{Fe}(+2), \text{Fe}(+4)]$ oxide
 - It is a non-stoichiometric compound
 - It is a mixed $[\text{Fe}(+2), \text{Fe}(+3)]$ oxide
 - Each Fe has $+8/3$ oxidation state
61. Equivalent weight of FeC_2O_4 in the change, $\text{FeC}_2\text{O}_4 \rightarrow \text{Fe}^{3+} + \text{CO}_2$ is
- M/3
 - M/6
 - M/2
 - M/1
62. For the cell prepared from electrode A and B. Electrode A : $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}$, $E_{\text{red}}^0 = +1.33 \text{ V}$ and Electrode B : $\text{Fe}^{3+}/\text{Fe}^{2+}$, $E_{\text{red}}^0 = 0.77 \text{ V}$ which of the following statements is not correct
- The electrons will flow from B to A when connection are made
 - The emf of the cell will be 0.56 V
 - A will be positive electrode
 - None of these
63. If $E_{\text{Fe}^{2+}/\text{Fe}}^0$ is x_1 , $E_{\text{Fe}^{3+}/\text{Fe}}^0$ is x_2 then $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0$ will be :
- $3x_2 - 2x_1$
 - $x_2 - x_1$
 - $2x_1 - 3x_2$
 - $2x_1 + 3x_2$
64. The hydrogen electrode is dipped in a solution of pH 3 at 25°C . The reduction potential of the electrode would be (the value of $\frac{2.303 RT}{F}$ is 0.059 V):
- 0.177 V
 - 0.177 V
 - 0.087 V
 - 0.059 V
65. If the solution of the CuSO_4 in which copper rod is immersed is diluted to 10 times, the reduction electrode potential
- increase by 0.030 volt
 - decrease by 0.030 volt
 - increase by 0.059 volt
 - decrease by 0.0059 volt
66. At pH = 2, $E_{\text{Quinhydrone}}$ will be ($E_{\text{Quinhydrone}}^0 = 1.30 \text{ V}$) -
-
- 1.36 V
 - 1.30 V
 - 1.42 V
 - 1.20 V
67. In which of the following cells net cell reaction is :
- $$\text{Ag}^+ (\text{aq}) + \text{Cl}^- (\text{aq}) \longrightarrow \text{AgCl} (\text{s})$$
- $\text{Ag} \mid \text{AgCl}(\text{s}) \mid \text{KCl} (\text{aq}) \parallel \text{AgNO}_3 \mid \text{Ag}$
 - $\text{Ag} \mid \text{AgCl}(\text{s}) \mid \text{KCl} (\text{aq}) \mid \text{Ag}$
 - $\text{Ag} \mid \text{AgNO}_3 \parallel \text{KCl} (\text{aq}) \mid \text{AgCl} \mid \text{Ag}$
 - All of the above
68. A standard hydrogen electrode has zero electrode potential because :
- hydrogen is easiest to oxidise
 - this electrode potential is assumed to be zero
 - hydrogen atom has only one electron
 - hydrogen is the lightest element
69. The following facts are available
- $$2\text{X}^- + \text{Y}_2 \rightarrow 2\text{Y}^- + \text{X}_2; \quad 2\text{W}^- + \text{Y}_2 \rightarrow \text{No reaction}$$
- $$2\text{Z}^- + \text{X}_2 \rightarrow 2\text{X}^- + \text{Z}_2$$
- Which of the following is correct ?
- $E_{\text{W}^-/\text{W}_2}^0 > E_{\text{Y}^-/\text{Y}_2}^0 > E_{\text{X}^-/\text{X}_2}^0 > E_{\text{Z}^-/\text{Z}_2}^0$
 - $E_{\text{W}^-/\text{W}_2}^0 < E_{\text{Y}^-/\text{Y}_2}^0 < E_{\text{X}^-/\text{X}_2}^0 < E_{\text{Z}^-/\text{Z}_2}^0$
 - $E_{\text{W}^-/\text{W}_2}^0 < E_{\text{Y}^-/\text{Y}_2}^0 > E_{\text{X}^-/\text{X}_2}^0 > E_{\text{Z}^-/\text{Z}_2}^0$
 - $E_{\text{W}^-/\text{W}_2}^0 > E_{\text{Y}^-/\text{Y}_2}^0 < E_{\text{X}^-/\text{X}_2}^0 < E_{\text{Z}^-/\text{Z}_2}^0$

70. Saturated solution of KNO_3 is used to make 'salt bridge' because -
 (1) Velocity of K^+ is greater than that of NO_3^-
 (2) Velocity of NO_3^- is greater than that of K^+
 (3) Velocities of both K^+ and NO_3^- are nearly the same
 (4) KNO_3 is highly soluble in water
71. Incorrect order of radius is :
 (1) $\text{N}^{3-} > \text{F}^- > \text{Na}^+$
 (2) $\text{P}^{3-} > \text{S}^{2-} > \text{K}^+$
 (3) $\text{S}^{2-} > \text{Cl}^- > \text{F}^-$
 (4) $\text{I} > \text{Ga} > \text{Al}$
72. Which of the following configuration has least I.P. ?
 (1) $1s^2, 2s^2 2p^6$
 (2) $1s^2, 2s^2 2p^6, 3s^2 3p^4$
 (3) $1s^2, 2s^2 2p^6, 3s^2 3p^3$
 (4) $1s^2, 2s^2 2p^5$
73. In which one of the following pairs the radius of the second species is greater than that of the first ?
 (1) $\text{O}^{2-}, \text{N}^{3-}$ (2) Na, Mg
 (3) Al, B (4) $\text{Li}^+, \text{Be}^{2+}$
74. Element of Atomic number 55 belongs to :-
 (1) s-block (2) p-block
 (3) d-block (4) f-block
75. The first 3-I.P.'s are 240, 830, 6440 KJ/mole respectively. The element is :-
 (1) Na (2) Ca (3) Al (4) B
76. A neutral atom A is converted to A^{+3} by the following process :-

$$\text{A} \xrightarrow[\text{E}_1]{-\text{le}^\ominus} \text{A}^+ \xrightarrow[\text{E}_2]{-\text{le}^\ominus} \text{A}^{+2} \xrightarrow[\text{E}_3]{-\text{le}^\ominus} \text{A}^{+3}$$

 The correct order of E_1 , E_2 and E_3 energies is :-
 (1) $\text{E}_1 < \text{E}_2 < \text{E}_3$ (2) $\text{E}_1 > \text{E}_2 > \text{E}_3$
 (3) $\text{E}_1 = \text{E}_2 = \text{E}_3$ (4) $\text{E}_1 > \text{E}_2 < \text{E}_3$
77. Increasing order of Z_{eff} is :-
 (1) $\text{K}^+ < \text{K}$ (2) $\text{Cl} < \text{Cl}^-$
 (3) $\text{Ca}^+ < \text{Ca}^{+2}$ (4) $\text{O}^- < \text{O}^{2-}$
78. Electronic configuration of elements A and B are :-
 $\text{A} = 2s^2 2p^3$ $\text{B} = 2s^2 2p^4$
 What is wrong about them :
 (1) IE $\text{A} > \text{B}$ (2) EA $\text{A} < \text{B}$
 (3) EN $\text{A} < \text{B}$ (4) Size $\text{A} < \text{B}$
79. Consider the 1st I.P. of the elements, which electronic configuration have maximum and minimum I.P. of among the following :-
 (A) $2s^2 2p^3$ (B) $2s^2 2p^4$
 (C) $2s^2 2p^1$ (D) $2s^2$
 (1) A and C (2) B and C
 (3) A and B (4) C and D
80. Relative decrease in vapour pressure is 0.4 for a solution containing 1 mole AB in 3 mol H_2O , therefore AB is ionised at the extent of
 (1) 60% (2) 50%
 (3) 100% (4) 40%
81. The vapour pressure of pure liquid solvent A is 0.80 atm. When a non-volatile substance B is added to the solvent, its vapour pressure drops to 0.60 atm. mol fraction of the component B in the solution is -
 (1) 0.50 (2) 0.25
 (3) 0.75 (4) 0.40
82. Solute A is a ternary electrolyte and solute B is a non-electrolyte. If 0.1 M solution of solute B produces an osmotic pressure of 2P, then 0.05 M solution of A at the same temperature will produce an osmotic pressure equal to -
 (1) 1.5 P (2) 2P (3) 3P (4) P

83. Study the following figure, and choose the correct options.



- (1) There will be no movement of any solution across the membrane
- (2) Solvent of BaCl_2 will flow towards the NaCl solution
- (3) NaCl will flow towards the BaCl_2 solution
- (4) The osmotic pressure of 0.1 M NaCl is lower than the osmotic pressure of 0.05 M BaCl_2 , assuming complete dissociation of the electrolyte
84. What is the molarity of H_2SO_4 solution that has a density of 1.84 gm/cc at 35°C and contains 98% by weight-
- (1) 4.18 M (2) 8.14 M
- (3) 18.4 M (4) 18 M
85. A 5.25% solution of a substance is isotonic with a 1.5% solution of urea (molar mass = 60 g mol^{-1}) in the same solvent. If the densities of both the solutions are assumed to be equal to 1.0 gcm^{-3} , molar mass of the substance will be-
- (1) 90.0g mol^{-1}
- (2) 115.0g mol^{-1}
- (3) 105.0g mol^{-1}
- (4) 210.0 g mol^{-1}
86. A real gas is expected to behave most ideally at -
- (1) low temperature and low pressure
- (2) low temperature and high pressure
- (3) high temperature and low pressure
- (4) high temperature and high pressure

87. Consider the following statements :

If the van der Waals parameters of two gases are given as :

$$a \text{ (atm L}^2\text{mol}^{-2}\text{)} \quad b \text{ (Lmol}^{-1}\text{)}$$

$$\text{Gas X :} \quad 6.5 \quad 0.056$$

$$\text{Gas Y :} \quad 8.0 \quad 0.011$$

$$\text{then} \quad A : V_C(X) < V_C(Y)$$

$$B : P_C(X) < P_C(Y)$$

$$C : T_C(X) < T_C(Y)$$

Select correct alternate :

- (1) A alone
- (2) A and B
- (3) A, B and C
- (4) B and C
88. At high temperature and low pressure, the van der Waals' equation is reduced to:

$$(1) \left(P + \frac{a}{V^2} \right) V = RT$$

$$(2) PV = RT$$

$$(3) P(V-b) = RT$$

$$(4) \left(P + \frac{a}{V^2} \right) (V-b) = RT$$

89. Select the correct statements about van der Waals' const 'b':

1. Its unit is mol litre^{-1}

2. It depends on intermolecular force

3. Its value depends on molecular size

(1) 2, 3 (2) 1, 2 (3) 1, 3 (4) 1,2,3

90. At a constant pressure what should be the percentage increase in temperature for a 10% increase in volume ?

(1) 10%

(2) 11%

(3) 8.9%

(4) 12.5%

91. How many plants in the list given below are the members of non vascular embryophytes :-
Spirogyra, Volvox, Fucus, Polysiphonia, Polytrichum, Sphagnum, Marchantia, Funaria, Selaginella, Equisetum.

- (1) Six (2) Three
(3) Four (4) Five

92. Match the column A with column B and find out the correct answer :-

Column-A

Column-B

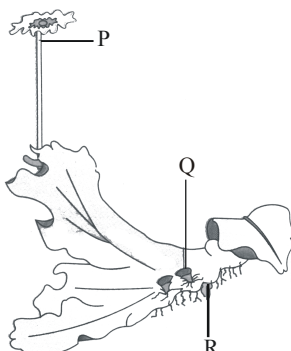
- (a) Bacteria (P) Soap box
(b) Blue green algae (Q) Mucilagenous sheath
(c) Dinoflagellates (R) Red tide
(d) Chrysophyte (S) Plasmid

- (1) a-S, b-R, c-P, d-Q
(2) a-S, b-R, c-Q, d-P
(3) a-Q, b-P, c-R, d-S
(4) a-S, b-Q, c-R, d-P

93. Which of the following statement is correct ?

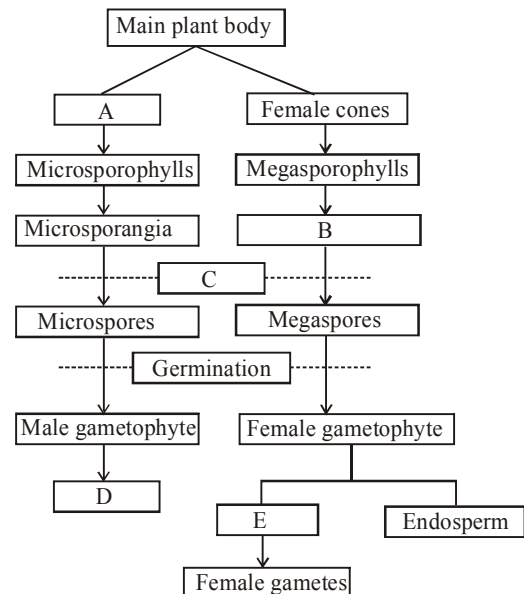
- (1) In mosses asexual reproduction takes place by green, multicellular buds called gemmae
(2) The sporophyte of bryophytes is differentiated in foot, seta and capsule and it is a free-living generation
(3) Protonema is a green, creeping and filamentous stage in mosses, which develops directly from a spore.
(4) Liverworts have an elaborate mechanism of seed dispersal.

94. In given figures, identify P, Q and R :-



	P	Q	R
(1)	Antheridiophore	Gemma cup	Rhizoids
(2)	Antheridal branch	Capsule	Archegoniophore
(3)	Archegoniophore	Capsule	Rhizoids
(4)	Antheridiophore	Gemma cup	Roots

95. The given flow chart shows the reproductive structure & events of a gymnospermic plant. Identify the structure/events those are blank.



	A	B	C	D	E
(1)	Male cone	Micro-sporangia	Meiosis	Archegonia	Antherozoids
(2)	Sori	Micro-sporangia	Meiosis	Antherozoids	Archegonia
(3)	Male cone	Mega-sporangia	Meiosis	Male gametes	Archegonia
(4)	Sori	Mega-sporangia	Mitosis	Male gametes	Egg cell

96. Which of the following statement is not correct ?

- (1) In gymnosperms the sporophyte and the gametophyte do not have an independent free-living existence.
(2) The leaves of gymnosperms are well-adapted to withstand extremes of temperature humidity and wind.
(3) Pollination in gymnosperms is mainly carried by air current

(4) The multicellular female gametophyte of gymnosperm is retained within megasporangium.

97. In which one of the following, the genus name, its character and its class are not correctly matched, whereas the remaining three are correctly matched ?

	Genus Name	Character	Class
(1)	<i>Alternaria</i>	Sexual stage is absent	Deuteromycetes
(2)	<i>Neurospora</i>	Endogenous ascospores	Ascomycetes
(3)	<i>Pythium</i>	Dikaryotic mycelium	Basidiomycetes
(4)	<i>Albugo</i>	Aseptate and coenocytic mycelium	Phycomycetes

98. Cyanophycean starch is a glycogen like material structurally related to:-

- (1) Chitin (2) Mannitol
(3) Laminarin (4) Amylopectin

99. Presence of rigid cell wall and formation of embryo are characterized by kingdom.....

- (1) Protista (2) Plantae
(3) Monera (4) Animalia

100. Which organisms live in extreme salty area?

- (1) Halophiles (2) Methanogens
(3) Spirochaete (4) Cyanobacteria

101. Rhodophyceae is red coloured due to

- (1) Xanthophyll (2) Carotenoids
(3) γ -phycoerythrin (4) γ -phycocyanin

102. Floridean starch is found in

- (1) Chlorophyceae (2) Rhodophyceae
(3) Myxophyceae (4) Cyanophages

103. Smallest plant which contain green pigment such as higher green plant is

- (1) Schizomycetes
(2) Rhodophyceae
(3) Chlorophyceae
(4) Phaeophyceae

104. In oogamy, fertilization involves

- (1) A large non-motile female gamete and a small non-motile male gamete
(2) A large motile female gamete and a small non-motile male gamete
(3) A small non-motile female gamete and a large motile male gamete
(4) A large non-motile female gamete and a small motile male gamete

105. Agar agar is obtained from

- (1) Green algae (2) Red algae
(3) Brown algae (4) Blue green algae

106. Which one of the following is a non-vascular embryophyte

- (1) Thallophyta (2) Bryophyta
(3) Pteridophyta (4) All the above

107. Bryophytes can be separated from algae, because they

- (1) Are thalloid forms
(2) Have no conducting tissue
(3) Possess archegonia
(4) Contain chloroplast

108. In bryophytes, antherozoids are

- (1) Biflagellate
(2) Multiflagellate
(3) Sometimes biflagellate and sometimes multiflagellate
(4) Biflagellate in a few species and multiflagellate in the rest

109. Identify the correct sequence of taxa in Linnaean hierarchy

- (1) Species, genus, family, order, class
(2) Class, family, species, genus, order
(3) Phylum, class, family, species, order
(4) Species, genus, phylum, family, class

110. In five kingdom classification land plants are classified under

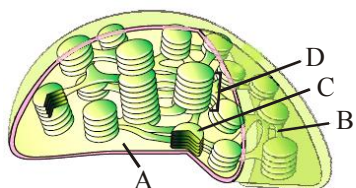
- (1) Bryophyta, Pteridophyta and spermatophytes
(2) Bryophyta and Algae
(3) Embryophyta and fungi
(4) Bryophyta and Bacteria

- 111.** Cyanobacteria are
- (1) Mosses which attack bacteria
 - (2) Bacteria which attack cyanophyceae
 - (3) Autotrophic organism with phycocyanin
 - (4) None of these
- 112.** Plasmids are
- (1) Viruses
 - (2) New micro-organisms
 - (3) Extra-chromosomal genetic material
 - (4) None of these
- 113.** Sex organs of thallophytes are generally
- (1) Unicellular
 - (2) Multicellular surrounded by a sterile jacket
 - (3) Multicellular but not surrounded by any jacket
 - (4) Unicellular with a sterile jacket
- 114.** Fern's prothallus normally is
- (1) Haploid (2) Diploid
 - (3) Triploid (4) Tetraploid
- 115.** Pteridophytes are characterized by
- (1) Presence of vascular system
 - (2) Absence of flowers
 - (3) Absence of fruits
 - (4) All the above
- 116.** A plant having vascular supply, producing spores but lacking seed is a
- (1) Byophyte (2) Pteridophyte
 - (3) Gymnosperm (4) Angiosperm
- 117.** Sporophyte dependent upon gametophyte is found in
- (1) Algae (2) Fungi
 - (3) Bryophytes (4) Pteridophytes
- 118.** Spore producing body of a cellular slime mould is
- (1) Pseudoplasmodium
 - (2) Plasmodium
 - (3) Sporangium
 - (4) Sporophore
- 119.** Which of the following statements is/are correct about cell theory ?
- (1) New cells are formed through meiosis of pre existing cells.
 - (2) Cell is only functional unit of organism.
 - (3) All living organisms are made up tissues and their products.
 - (4) Cell is basic unit of organisms.
- 120.** (1) In plasma membrane, lipids are arranged in two layers.
- (2) Each lipid molecule possess a Nonpolar hydrophilic head towards outer surface.
- (3) Proteins may be peripheral or integral.
- (4) Tail of lipid is hydrophilic.
- Of the above, how many statements are true ?
- (1) One (2) Two
 - (3) Three (4) Four
- 121.** $\text{Na}^+ - \text{K}^+$ pump is an example of :
- (1) Active transport
 - (2) Passive transport
 - (3) Ionic transport
 - (4) Carrer protein independent transport
- 122.** (I) Endoplasmic reticulum
(II) Chloroplast
(III) Golgi body
(IV) Lysosome
(V) Mitochondria
(VI) Vacuoles
(VII) Ribosome
- Of the above, which set of organelles are considered as endomembrane system ?
- (1) I, II, III, IV (2) I, III, IV, VI
 - (3) I, V, VI, VII (4) II, V, VI, VII
- 123.** Substances synthesized with in ER are packed into through and then released into
- (1) Vacuoles, golgi complex, Lysosome
 - (2) Lysosome, golgi complex, Cytoplasm
 - (3) Vesicles, golgi complex, Cytoplasm
 - (4) Vesicles, Lysosome, Cytoplasm

124. Which of the following statements is/are true for lysosome ?

- (1) They are important in phagocytosis and pinocytosis like process
- (2) They are responsible for degradation of only living cells.
- (3) It consists of hydrolases type enzymes (Lipases, transferases, ligases).
- (4) All of the above

125.



Identify the labelled part A, B, C and D in the above figure of chloroplast.

	A	B	C	D
(1)	Stroma	Stroma lamella	Thylakoid	Granum
(2)	Cytosol	Stroma lamella	Thylakoid	Granum
(3)	Cytoplasm	Granum	Thylakoid	Stroma
(4)	Stroma	Thylakoid	Granum	Stroma lamella

126. The lipids are arranged within the membrane with the polar head towards the :-

- (1) Inner sides
- (2) Outer sides
- (3) Intrinsic proteins
- (4) All sides

127. Find out the **correct** match from the following table.

	Column-I	Column-II	Column-III
(i)	Mitochondria	Cristae	Oxidative phosphorylation
(ii)	Lysosome	Single membrane bound	Protein synthesis
(iii)	Chloroplast	Thylakoids	Photo phosphorylation

- (1) (i), (ii)
- (2) (i), (iii)
- (3) (iii)
- (4) (i), (ii), (iii)

128. Which one of the following cellular part is correctly described ?

- (1) Ribosomes, those in chloroplasts are 80-S type
- (2) Lysosomes optimally active at a pH of about 8.1
- (3) Thylakoids have system for ATP formation
- (4) Centrioles-sites for RNA synthesis

129. Read the following five statements A-E and answer as asked next to them

A : Chlorophyll and carotenoids are responsible for trapping light energy essential for protein synthesis.

B : Elaioplasts store oil and fats.

C : The mitochondria divides by fission.

D : The vacuole is the membrane less space found in the cytoplasm of plant cell.

E : Golgi apparatus found in both plant and animal cells.

How many of the above statements are correct?

- (1) Two
- (2) Three
- (3) Four
- (4) One

130. Which statement is wrong ?

- (1) Cytoskeleton maintain the shape of cell
- (2) Two centrioles presents in centrosome
- (3) 80-s ribosomes founds only in plant cells
- (4) A green alga Chlamydomonas contains one chloroplast

131. Which pair is wrong

Item-A

Item-B

- (1) Golgi body Formation of glycoproteins
- (2) Thylakoid Liberation of oxygen
- (3) RER Ribosome formation
- (4) Golgi apparatus Cell plate formation

132. Acidic hydrolases presents in animal cells

- (1) Peroxisomes
- (2) Glyoxisomes
- (3) Lysosomes
- (4) Vacuole

133. If in a cell, the amount of DNA is $2C$ in G_2 phase then what will be the ratio of amount of DNA in metaphase : G_1 phase : Anaphase :-

- (1) $2 : 1 : 2$ (2) $1 : 2 : 2$
(3) $2 : 1 : 4$ (4) $1 : 1 : 2$

134. Single chromatid chromosomes can be seen during :-

- (1) Prophase, Anaphase - I, Telophase - I
(2) Anaphase, Anaphase - II
(3) Metaphase, Metaphase - II, Anaphase - II
(4) Anaphase - I, Anaphase

135. The formation of bivalent and synaptonemal complex is a feature of :-

- (1) Metaphase-I (2) Zygotene
(3) Pachytene (4) Diakinesis

136. Meiosis involves :-

- (1) One division of nucleus and one division of chromosomes
(2) One division of nucleus and two division of chromosomes
(3) Two division of nucleus and two division of chromosomes
(4) Two division of nucleus and one division of chromosomes

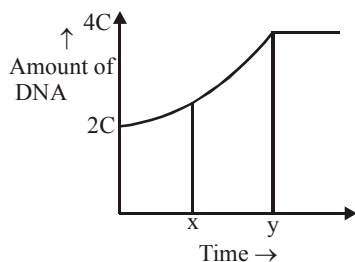
137. The stage which is characterised by the appearance of recombination nodules is :-

- (1) Diplotene (2) Pachytene
(3) Zygotene (4) Leptotene

138. If in a cell 12 tetrad are present in prophase-I then what will be the number of chromosomes in each cell in anaphase-II ?

- (1) 24 (2) 6 (3) 12 (4) 48

139. The given graph shows the change in DNA content in a single cell during the mitosis which stage occurs between the time x and y ?

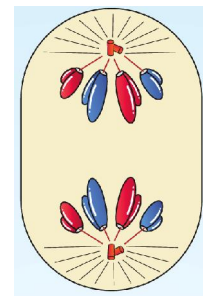


- (1) Interphase (2) Prophase
(3) Anaphase (4) Telophase

140. In onion root tip cell there are 14 chromosomes and the DNA content is $2C$ after M-phase, what will be the number of chromosome and amount of DNA at G_1 , after S and G_2 phase respectively ?

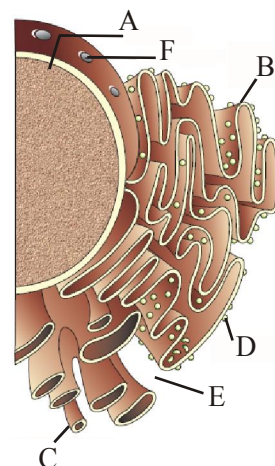
	No. of chromosomes	Amount of DNA
(1)	14, 14, 14	$2C$, $4C$, $4C$
(2)	14, 28, 28	$2C$, $2C$, $4C$
(3)	14, 14, 28	$4C$, $4C$, $4C$
(4)	28, 14, 14	$4C$, $2C$, $2C$

141. A stage of cell division is shown in the figure. Select the correct match :-



(1)	Metaphase	Nuclear membrane disintegration
(2)	Pachytene	Crossing over in homologous chromosomes.
(3)	Anaphase	Split of centromere, movement of daughter chromosomes
(4)	Diakinesis	Terminalisation of chromosomes

142. Identify the components labelled A, B, C, D, E and F in the diagram below from the list (i) to (vi) given along with



Components :-

- | | |
|------------------|----------------|
| (i) SER | (ii) Ribosome |
| (iii) Nucleus | (iv) Cytoplasm |
| (v) Nuclear pore | (vi) RER |

The correct components are

- | | | | | | |
|---------|-----|----|----|----|-----|
| A | B | C | D | E | F |
| (1) v | i | vi | iv | ii | iii |
| (2) i | iii | ii | vi | v | iv |
| (3) iii | vi | i | ii | iv | v |
| (4) iii | i | ii | vi | v | iv |

- 143.** An elaborate network of filamentous proteinaceous structures present in the cytoplasm which helps in the maintenance of cell shape is called:-

- | | |
|---------------|------------------|
| (1) ER | (2) Plasmalemma |
| (3) Thylacoid | (4) Cytoskeleton |

- 144.** Condensation of chromosomes occurs in

- | | |
|-----------------|----------------|
| (1) Anaphase | (2) Prophase-I |
| (3) Anaphase-II | (4) Telophase |

- 145.** The number of DNA in chromosomes at the G₂-Stage of cell cycle in both mitosis and meiosis is :

- | | |
|--------------------|--------------------|
| (1) Remains same | (2) Becomes half |
| (3) Doubles itself | (4) Triples Itself |

- 146.** Assertion (A) : Meiotic division result in the production of haploid cells.

Reason (R) : In zygotene stage of mitosis, synaptonemal complex formed.

- | |
|---|
| (1) A & R are true, R is correct explanation. |
| (2) A & R true, R is not correct explanation. |
| (3) A is true, R is false. |
| (4) A is false, R is true. |

- 147.** A bivalent consists of

- | |
|---|
| (1) Two chromatids and one centromere |
| (2) Four chromatids and two centromere |
| (3) Two chromatids and two centromere |
| (4) Four chromatids and four centromere |

- 148.** Which of the following organisms are unicellular ?

- | | |
|--------------------------|----------------------|
| (1) <i>Paramecium</i> | (2) Yeast |
| (3) <i>Chlamydomonas</i> | (4) All of the above |

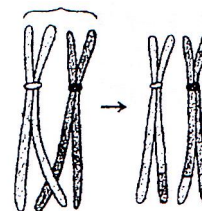
- 149.** Find out correct match pair

- | |
|---|
| (1) Mitochondria – Dark reaction |
| (2) Chloroplast – light reaction |
| (3) Thylakoid – kreb's cycle |
| (4) Granum – Group of branched membranous structure |

- 150.** 70s ribosome are found in :

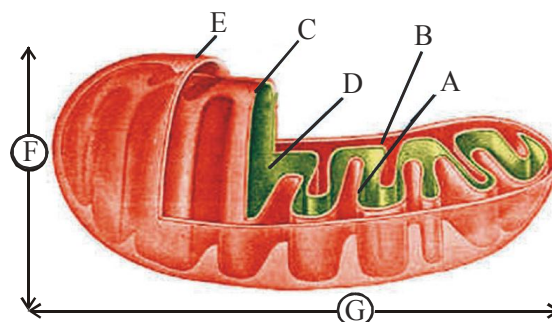
- | |
|--|
| (1) Prokaryotic cells and cytoplasm of eukaryotic cells. |
| (2) Mitochondria, Chloroplasts and prokaryotic cells. |
| (3) Mitochondria, ER and prokaryotic cells. |
| (4) Mitochondria ER and eukaryotic cells. |

- 151.** Given below is the representation of a certain event at particular stage of a type of cell division. Which is this stage ?



- | |
|--|
| (1) Prophase of Mitosis |
| (2) Both prophase and metaphase of mitosis |
| (3) Prophase I during meiosis |
| (4) Prophase II during meiosis |

- 152.** Match the columns



Column I		Column II	
a.	Labelled-A	i.	Inner membrane
b.	Labelled-B	ii.	Outer membrane
c.	Labelled-C	iii.	Inter membrane space
d.	Labelled-D	iv.	Crista
e.	Labelled-E	v.	Matrix
		vi.	Inner chamber
		vii.	Cisternae

- (1) a(iv), b(iii), c(i), d(v), e(ii)
- (2) a(vii), b(vi), c(i), d(iii), e(ii)
- (3) a(ii), b(iii), c(vii), d(v), e(iv)
- (4) a(i), b(ii), c(vi), d(v), e(iii)

153. Which cell organelle divides the intracellular space into two distinct compartments, i.e., luminal (inside) and extra luminal (cytoplasm) compartments

- (1) Golgibody (2) Mitochondria
- (3) ER (4) Lysosome

154. Choose the option with correct sequence of events at the neuromuscular junction and during muscle contraction :-

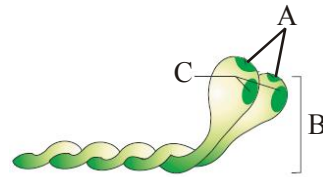
- (a) Binding of Ca^{2+} with troponin and removal of masking of active sites for myosin on actin filaments
- (b) Action potential in the motor neuron and release of neurotransmitter
- (c) Depolarisation of sarcolemma
- (d) Formation of cross-bridges
- (e) Release of Ca^{2+} from sarcoplasmic reticulum by passive transport
- (f) Sliding of actin filaments along with Z-line towards the centre of the A-band.

- (1) a, b, c, d, e, f
- (2) a, b, c, e, d, f
- (3) b, c, e, a, d, f
- (4) b, c, a, d, e, f

155. Which one of the following is not associated with thick filaments [myosin] of myofibril

- (1) Tropomyosin
- (2) Meromyosin
- (3) ATPase enzyme
- (4) Cross arm

156.



The above figure is related with myosin monomer (Meromyosin). Identify A to C-

- (1) A-head, B-LMM, C-GTP binding sites
- (2) A-head, B-cross arm, C- Ca^{+2} binding sites
- (3) A-Actin binding site, B-cross arm, C-ATP binding sites
- (4) A-myosin binding site, B-HMM, C-ATP binding sites

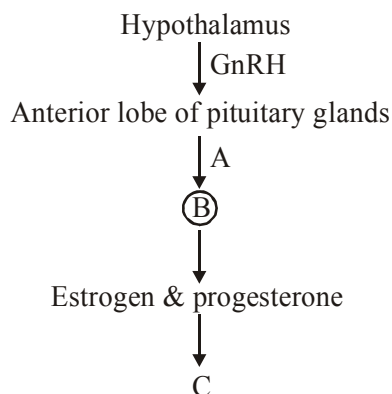
157. Select the incorrectly matched pair from the following:-

- (1) **Myasthenia gravis** : Auto immune disorder affecting myofibrils leading to weakening of skeletal muscles
- (2) **Muscular dystrophy** : Progressive degeneration of skeletal muscle mostly due to genetic disorder.
- (3) **Tetany** : Rapid spasms in muscle due to low Ca^{2+} in body fluids
- (4) **Muscular fatigue**: Due to lactic acid accumulation, muscle contraction is stopped

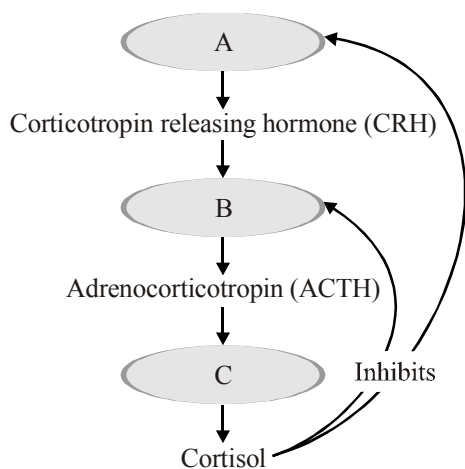
158. Which one of the following pair is not correctly matched ?

- (1) Pineal body : Melatonin
- (2) Leydig cells : Androgen
- (3) Intestine : Secretin
- (4) Neurohypophysis : Oxytocin

- 159.** Flow chart given below showing the influence of hormones in females. Mark the A, B, C in correct sequence to complete the flow chart :-



- (1) Prolactin, mammary gland, ovary
 (2) FSH, ovary, Uterus
 (3) LH, uterus, mammary gland
 (4) FSH, corpus Luteum, ovary
- 160.** Which glands are indicated as A, B and C in the given figure ?



- (1) A-Hypothalamus B-Anterior pituitary
 C-Adrenal cortex
 (2) A-Hypothalamus B-Anterior pituitary
 C-Adrenal medulla
 (3) A-Pituitary B-Thyroid C-Pineal
 (4) A-Pituitary B-Thyroid C-Parathyroid
- 161.** Mark antagonistic hormones :-
- (1) Insulin and glucagon
 (2) Adrenaline and noradrenalin
 (3) Calcitonin and parathormone
 (4) Both (1) & (3)

- 162.** When blood volume reduces and osmotic concentration of blood increases, the osmoreceptors of hypothalamus becomes active to promote thirst and secretion of all the following hormones increases in our body except one :-

- (1) Renin (2) ADH
 (3) ANF (4) Aldosterone

- 163.** Calcium and phosphorus metabolism is disturbed due to the malfunctioning of :

- (1) Parathyroid (2) Pancreas
 (3) Salivary glands (4) Both (2) and (3)

- 164.**A..... Induces ovulation of fully mature follicles and maintains theB..... . Formed from remnants of the graafian follicles?

A

B

- | | |
|----------------|-----------------|
| (1) FSH | Corpus luteum |
| (2) L.H. | Corpus albicans |
| (3) L.H. | Corpus luteum |
| (4) 1 & 2 both | |

- 165.** Hypothyroidism during pregnancy causes defective development and maturation of the growing baby leading to-

- (1) Cretinism (2) Mental retardation
 (3) deaf-mutism (4) All of the above

- 166.** Heterospory is found in

- (1) *Selaginella* (2) *Salvinia*
 (3) *Marselia* (4) All the above

- 167.** In which of the following plant diploid stage generally dominates

- (1) Algae (2) Ferns
 (3) Mosses (4) Fungi

- 168.** Common characteristic between bryophytes and pteridophytes is

- (1) Vascularisation
 (2) Terrestrial habit
 (3) Water for fertilization
 (4) Independent sporophyte

169. Example rich in proteins and used as food by space travellers is

- (1) chlorella (2) spirulina
(3) both (4) laminaria

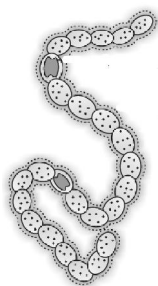
170. Branched chain lipids occur in the cell membranes of

- (1) Archaeobacteria (2) Mycoplasma
(3) Actinomycetes (4) Streptomyces

171. Some bacteria are not easily killed because of

- (1) Chitinous wall
(2) Endospore formation
(3) Presence of mesosome
(4) High tolerance

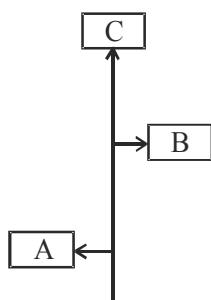
172. The given figure is representing



- (1) filamentous green algae
(2) filamentous yellow green algae
(3) filamentous Actinomycetes
(4) filamentous BGA (*Nostoc*)

173. Evolutionary relationship of three major group of organisms is shown in the accompanying diagram, where A, B, C are Archaeobacteria, Eubacteria and Cyanobacteria respectively. The *Nostoc* will be placed :

- (1) On branch A
(2) On branch B
(3) On branch C
(4) On branch A or on branch C



174. Some characters are given below about the archaebacteria which characters are not true about them :-

- (1) They live in most harsh habitats like salty areas hot springs and marshy areas
(2) They are different from other bacteria in having different structure of cell organelles like ER, GB
(3) Methanogens are present in guts of several ruminant animals like cows, buffaloes
(4) Due to their cell membrane they are able to survive in extreme conditions.

175. Which structure in bacteria acts as lysosomes of eukaryotes ?

- (1) Chromatophores
(2) Ribosomes
(3) Periplasmic space
(4) Mesosomes

176. Mycelium is aseptate and coenocytic in :-

- (1) Ascomycetes
(2) Basidiomycetes
(3) Deuteromycetes
(4) Phycomycetes

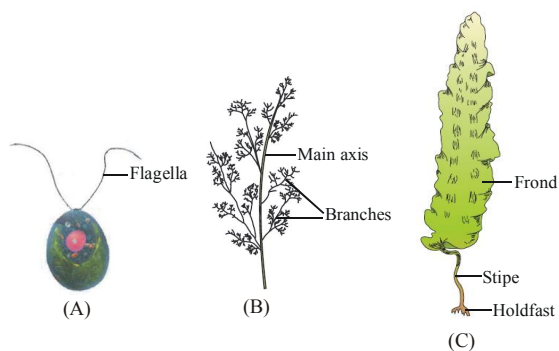
177. Which of the following is an example of slime mould ?

- (1) *Physarum* (2) *Fusarium*
(3) *Mucor* (4) *Usnea*

178. The sexual cycle in fungi involves certain steps. Which of the following represents **correct** sequence of steps ?

- (1) Plasmogamy → Meiosis → Karyogamy
(2) Karyogamy → Plasmogamy → Meiosis
(3) Plasmogamy → Karyogamy → Meiosis
(4) Meiosis → Plasmogamy → Karyogamy

179.



In the diagram given above, some of the algae have been labelled as 'A', 'B' and 'C'. These algae are respectively **correctly** identified as:-

- (1) *Fucus*, *Dictyota*, *Porphyra*
- (2) *Volvox*, *Porphyra*, *Chara*
- (3) *Chlamydomonas*, *Polysiphonia*, *Laminaria*
- (4) *Chlamydomonas*, *Laminaria*, *Porphyra*

180. Study the given diagram and give answer of questions a, b and c.



- (a) Name of plant ?
- (b) Class of plant ?
- (c) Ploidy of rhizoids ?

	(a)	(b)	(c)
(1)	<i>Sphagnum</i>	Lycopsida	Haploid
(2)	<i>Funaria</i>	Moss	Haploid
(3)	<i>Funaria</i>	Liverwort	Diploid
(4)	<i>Marchantia</i>	Moss	Haploid